QCD $u\bar{u} \to Zggg$: five reducers

Descriptive timings on a shared host. This $2 \to 4$ study requests all five reducers; TIRust times out before READY. There is no CPU reservation or formal performance-acceptance claim. The preceding 44 pages are preserved.

GNU Fortran (GCC) 15.2.0; AMD EPYC 9754 128-Core Processor; affinity CPU 12; one-thread settings; shared, unreserved host. Exact flags/runtime hashes are retained.

CLI generation: `loop_sm; u u~ > z g g g [virt=QCD]`. Each traversal uses all **10 saved points** at $\sqrt{s} = 1000$ GeV. This pilot uses the unselected first ten of a 100-point fixture frozen before amplitudes; no failed point is removed. The main campaign remains separate. Ten frozen Latin/reversed cycles skip unavailable TIRust slots; four live positions are near-balanced. Two calibrations fix one full traversal per timed batch.

Table 1: Warmed full-amplitude wall time: median ± unscaled MAD. Ratios use matched cycles relative to NinjaRS8; bracketed 95% intervals are descriptive.

Reducer	READY s	Warmed ms/point median ± MAD	Ratio / 8	Accepted timed backend:precision calls	3e-6 / 1e-8
CutTools	41.65	3708 ± 118.6	×1.056 [1.023, 1.093]	1:DP=100	pass / pass
Ninja	36.91	3490 ± 81.71	×0.9715 [0.9505, 1.017]	6:DP=100	pass / miss
COLLIER	38.91	3950 ± 114.0	×1.111 [1.070, 1.127]	7:DP=100	pass / miss
NinjaRS	35.36	3564 ± 63.52	×1.000 [1.000, 1.000]	8:DP=100	pass / miss
TIRust	600 s timeout	unavailable	—	—	unqualified

Fortran SYSTEM_CLOCK wall time is divided by actual points × loops. The minimum whole-cohort batch exceeds the nominal 1.1 s target; no points are thinned. 95% intervals use 10,000 percentile bootstrap resamples of the median matched-cycle ratio (seed 20261112), resampling ten whole cycles; they do not test baseline acceptance. Actual timed batches span 33.64–41.91 s. READY includes the first amplitude and warmup, excluding startup and SETPARA. TIRust was cleanly terminated at its original 600 s READY budget; its initialization and warmed timings are unavailable.

Numerical results and precision

All four completed lanes pass MadLoop's 3e-6 regression criterion against same-source CutTools DP. Reducers 6/7/8 miss the separate 1e-8 target in finite terms at points 1, 2, 4, 5; the maximum raw relative residual is 4.2473e-08 at point 4. Overall/sole-order duplicate rows are not extra physical points. TIRust's incomplete ten-point lane is unqualified. Its point-4 QP finite value clusters with 6/7/8 (spread 4.765e-12); this neither supplies a timing nor establishes absolute truth or fault. Every miss and the original timeout remain retained.

All four timed lanes have native-DP accepted outputs. TIRust's partial warmup has 4 calls (9:DP=1, 9:QP=3); these are neither timed counts nor a complete ten-point traversal. Accepted counts do not expose internal retries or exception flags. Cards use full finite+UV+IR (LoopPoleMode=0), explicit COLLIER poles and normal stability/rescue. Four READY lanes have live DSO-map witnesses. TIRust9 has executable/loader/ldd/shared-file binding, with no startup live-map claim.

Hua-Sheng's local TODO remains: source-matched baseline/rescue-chain comparisons and upper-95% paired runtime-ratio bound ≤1.05 for every main process.

TIRust source: `10ce34ed8ad5b6f5f989a0e220eb3a244a71d1a1`; DSO SHA-256 (runtime scope described above): `85ee0cdf860e5ad4c7a5c26163802673de22404690a10c7c53fb73ebe1ff5bb0`. Original closed capsule: experiments/zggg-five-backend-timing-1; index SHA-256 e401cf5d4efb7708c277172eaa3be9f29bcadf54c5b3b484e7311109cf9a26cb. Reproduce with source binding, pure derivation and CSV. CSV retains full precision. Style: pyAmpliCol `7266c1600f6e…`; complete revision and attribution are in style-provenance.json.